

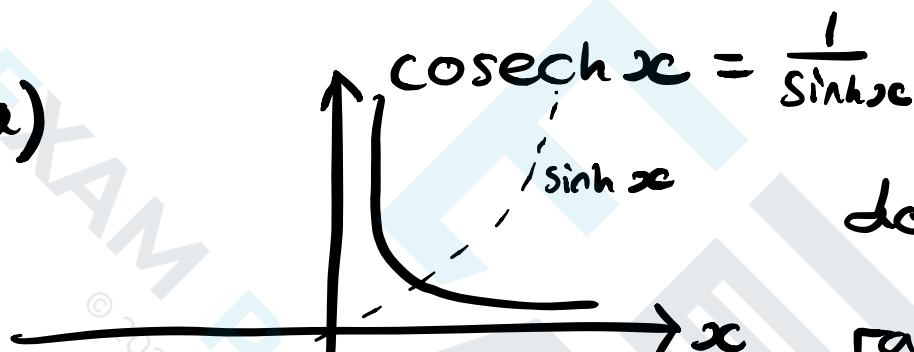
# Reciprocal hyperbolic trig functions

## Basic skills

Q1.  $\operatorname{sech} 2 = \frac{2}{e^2 + e^{-2}}$

$$\operatorname{sech} x = \frac{2}{e^x + e^{-x}}$$

Q2. a)

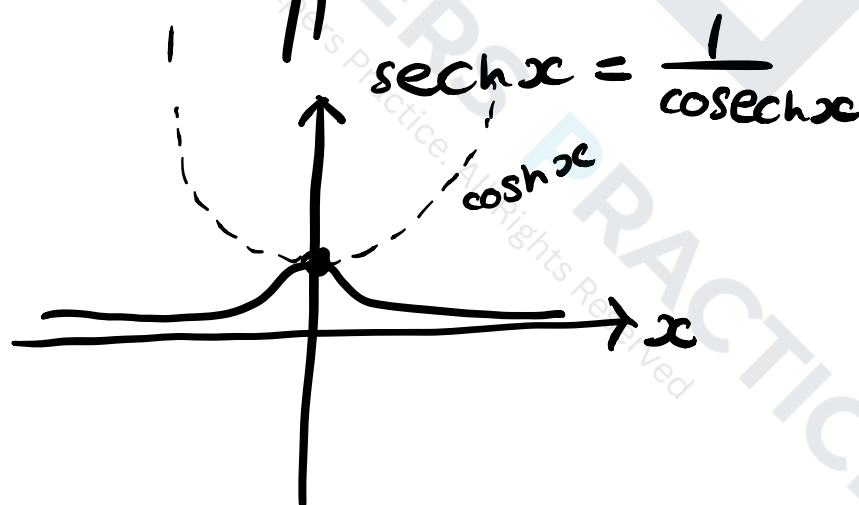


domain:  $x \neq 0$

range:  $y \neq 0$

← Asymptote at  $x = 0$

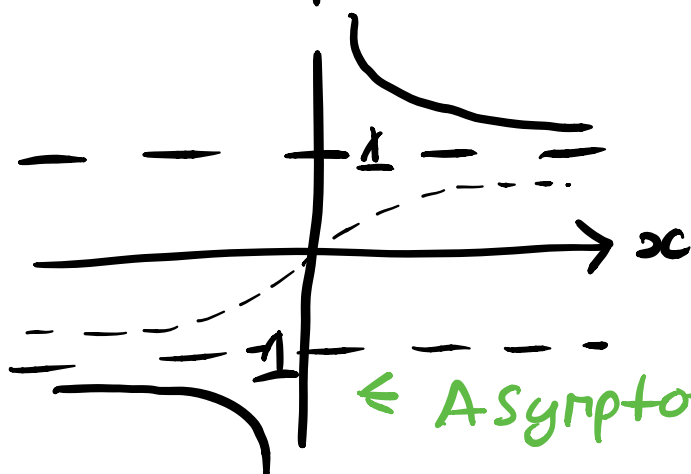
b)



domain:  $x \in \mathbb{R}$

range:  $0 < y \leq 1$

c)



domain:  $x \neq 0$

range:  $y > 1$  or  $y < -1$   
( $|y| > 1$ )

← Asymptote at  $x = 0$

Q3.

a) use  $\sinh^{-1}x = \ln(x + \sqrt{x^2+1})$

so  $\operatorname{cosech}^{-1}x = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2}+1}\right)$

b) use  $\cosh^{-1}x = \ln(x \pm \sqrt{x^2-1})$

so  $\operatorname{sech}^{-1}x = \ln\left(\frac{1}{x} \pm \sqrt{\frac{1}{x^2}-1}\right)$

c) use  $\tanh^{-1}x = \frac{1}{2} \ln\left(\frac{1-x}{1+x}\right)$

so  $\operatorname{coth}^{-1}x = \frac{1}{2} \ln\left(\frac{1+\frac{1}{x}}{1-\frac{1}{x}}\right) = \frac{1}{2} \ln\left(\frac{x+1}{x-1}\right)$

### Competency skills

Q4. a)  $\operatorname{sech}x = \frac{1}{2} \Rightarrow \frac{2}{e^x + e^{-x}} = \frac{1}{2}$

so  $e^x + e^{-x} = 4 \Rightarrow e^{2x} - 4e^x + 1 = 0$

$\Rightarrow e^x = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$

so  $x = \ln(2 \pm \sqrt{3})$

both positive

b)  $\operatorname{cosech}x = -4 \Rightarrow \frac{2}{e^x - e^{-x}} = -4$

$$\text{so } e^x - e^{-x} = -\frac{1}{2} \Rightarrow 2e^x - 2e^{-x} = -1$$

$$\Rightarrow 2e^{2x} + e^x - 2 = 0$$

$$\text{so } e^x = \frac{-1 \pm \sqrt{1 - 4 \times 2 \times -2}}{2 \times 2} = \frac{-1 \pm \sqrt{17}}{4}$$

now only  $\frac{-1 + \sqrt{17}}{4}$  positive so

$$x = \ln\left(\frac{-1 + \sqrt{17}}{4}\right)$$

$$c) \coth x = 2 \Rightarrow \frac{e^x + e^{-x}}{e^x - e^{-x}} = 2$$

$$\Rightarrow e^x + e^{-x} = 2e^x - 2e^{-x}$$

$$\Rightarrow e^x - 3e^{-x} = 0 \Rightarrow e^{2x} - 3 = 0$$

$$\text{so } e^{2x} = 3 \Rightarrow 2x = \ln 3 \text{ so } x = \frac{1}{2} \ln 3$$

$$\text{Q5. a) } \operatorname{sech} x = \frac{1}{2} \quad x = \operatorname{arcsech} \frac{1}{2}$$

$$\text{so } x = \ln\left(\frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2 - 1}\right) = \ln(2 \pm \sqrt{3})$$

$$b) \operatorname{cosech} x = -4 \quad x = \operatorname{arc cosech}(-4)$$

$$\text{so } x = \ln\left(-\frac{1}{4} + \sqrt{\left(-\frac{1}{4}\right)^2 + 1}\right) = \ln\left(-\frac{1}{4} + \frac{\sqrt{17}}{4}\right)$$

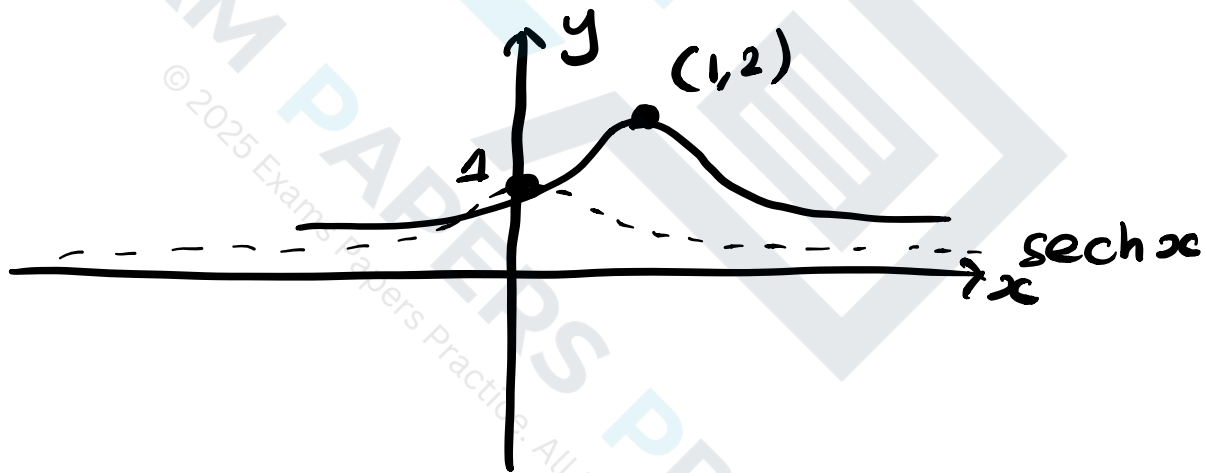
c)  $\coth x = 2 \quad x = \operatorname{arccoth} 2$

so  $x = \frac{1}{2} \ln \left( \frac{2+1}{2-1} \right) = \frac{1}{2} \ln 3$

All the same as in Q4.

Q6.  $y = 2 \operatorname{sech}(x-1)$

translate  $[0]$  stretch  $y$  by SF: 2



Q7.

$$\coth^2(2x^2) = (\coth(2x^2))^2$$

$$= \left( \frac{e^{2x^2} + e^{-2x^2}}{e^{2x^2} - e^{-2x^2}} \right)^2 = \frac{e^{4x^2} + 2 + e^{-4x^2}}{e^{4x^2} - 2 + e^{-4x^2}}$$

Q8.

$$\tanh^2 x = \left( \frac{e^x - e^{-x}}{e^x + e^{-x}} \right)^2 = \frac{e^{2x} - 2 + e^{-2x}}{e^{2x} + 2 + e^{-2x}}$$

$$= \frac{e^{2x} + 2 + e^{-2x} - 4}{e^{2x} + 2 + e^{-2x}} = 1 - \frac{4}{(e^x + e^{-x})^2} = 1 - \operatorname{sech}^2 x$$

Q9.

$$5 \operatorname{cosech} x - \coth x = 10$$

$$\text{so } \frac{10}{e^x - e^{-x}} - \frac{e^x + e^{-x}}{e^x - e^{-x}} = 10$$

$$\Rightarrow 10 - e^x - e^{-x} = 10e^x - 10e^{-x}$$

$$\Rightarrow 11e^x - 9e^{-x} - 10 = 0$$

$$\Rightarrow 11e^{2x} - 10e^x - 9 = 0$$

$$\text{so } e^x = \frac{10 \pm \sqrt{10^2 - 4 \times 11 \times -9}}{2 \times 11} = \frac{10 \pm 4\sqrt{31}}{22}$$
$$= \frac{5 \pm 2\sqrt{31}}{11}$$

$$\text{As } 5 - 2\sqrt{31} < 0 \quad \text{only } e^x = \frac{5 + 2\sqrt{31}}{11}$$

$$\text{so } x = \ln \left( \frac{5 + 2\sqrt{31}}{11} \right)$$

### Advanced skills

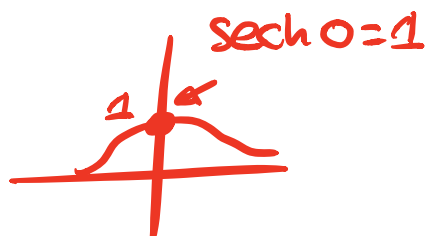
Q10.  $\tanh x \operatorname{cosech} x = 1$

$$\Rightarrow \frac{\sinh x}{\cosh x} \times \frac{1}{\sinh x} = 1 \Rightarrow \frac{1}{\cosh x} = 1$$

So  $\operatorname{sech} x = 1 \Rightarrow x = \operatorname{sech}^{-1} 1$

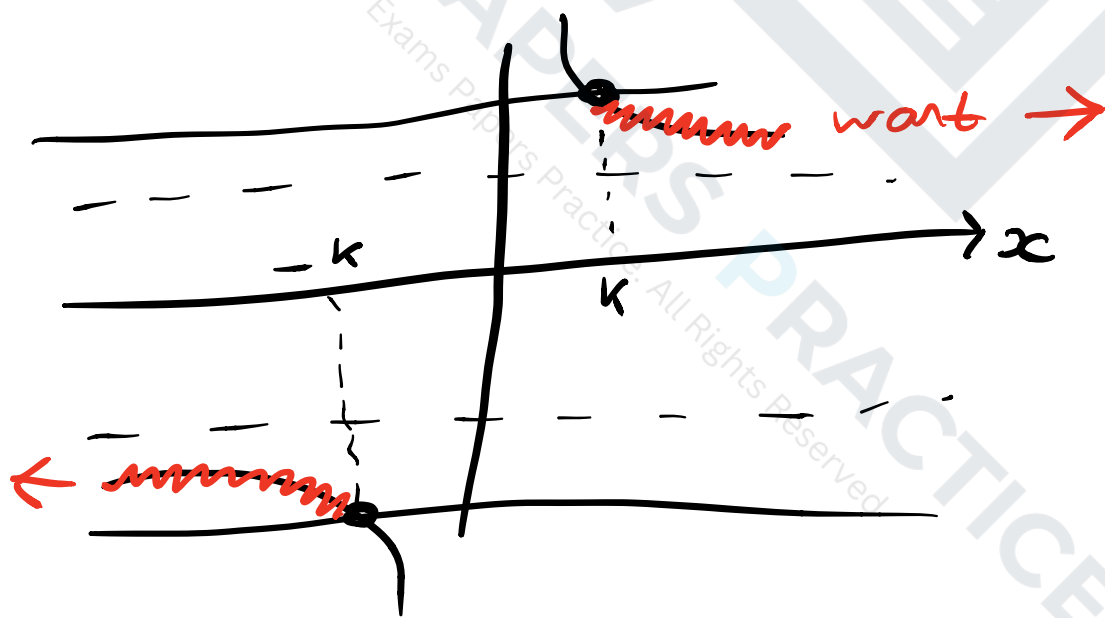
$$\Rightarrow x = \ln \left( \frac{1}{2} \pm \sqrt{\frac{1}{4} - 1} \right)$$

$$x = \ln(1) = 0$$

Could see this from graph 

only one solution as at peak

Q11.  $|\operatorname{coth} x| = 2 \quad \operatorname{coth} x = \pm 2$



So  $x = \operatorname{arccoth}(\pm 2)$

$$= \frac{1}{2} \ln \left( \frac{\pm 2 + 1}{\pm 2 - 1} \right) = \frac{1}{2} \ln 3, \frac{1}{2} \ln \left( \frac{1}{3} \right)$$

$$= \frac{1}{2} \ln 3, -\frac{1}{2} \ln 3$$

So  $k = \frac{1}{2} \ln 3$

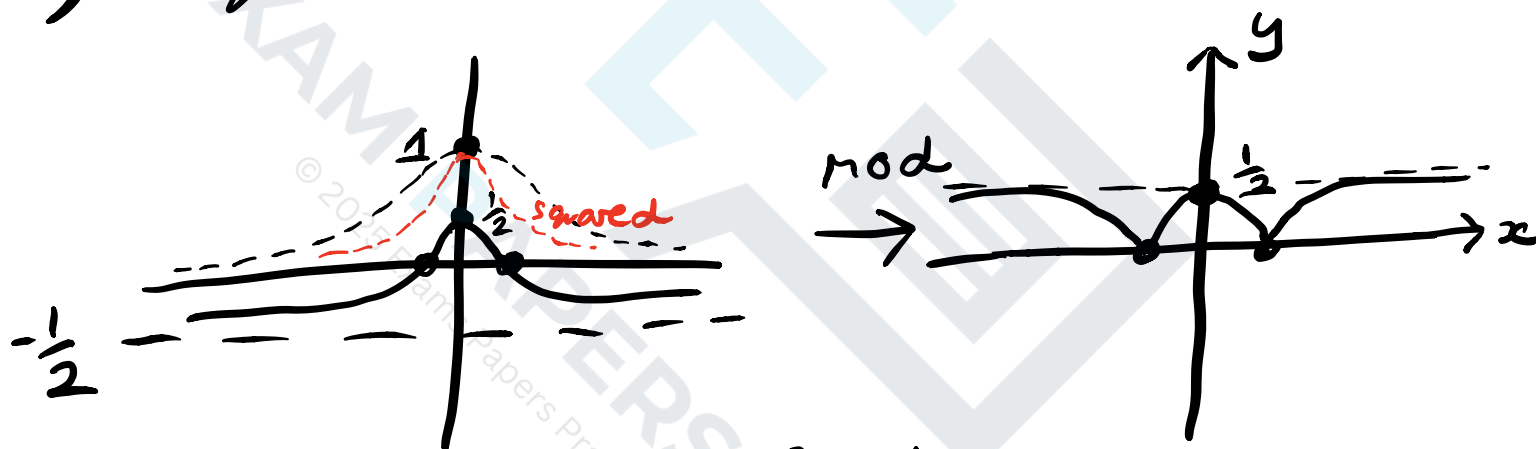
Q12.

$$\lim_{x \rightarrow \infty} \frac{\tanh x}{\sinh x} = \lim_{x \rightarrow \infty} \frac{1}{\cosh x} = 0$$

as  $\cosh x \rightarrow \infty$

Q13.  $y = |\operatorname{sech}^2 x - \frac{1}{2}|$

a) square then translate then modulus

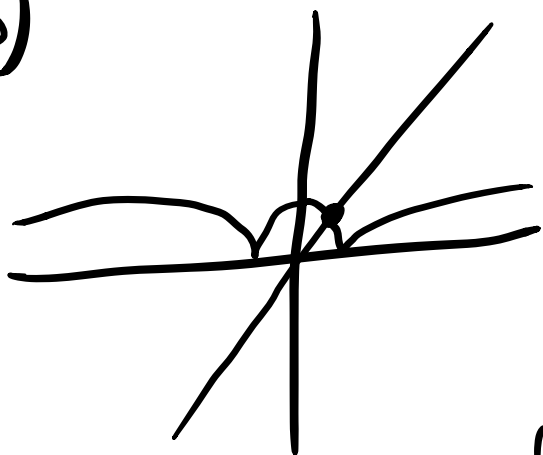


For  $x$  int when  $\operatorname{sech}^2 x = \frac{1}{2}$

$$\text{so } \operatorname{sech} x = \frac{1}{2} \Rightarrow x = \ln(4 \pm \sqrt{4^2 - 1})$$

$$\text{or } \cosh x = 4 \quad x = \ln(4 \pm \sqrt{15})$$

b)



so can ignore modulus

$$\operatorname{sech}^2 x - \frac{1}{2} = x$$

$$\text{so } 0 < x < \ln(4 + \sqrt{15})$$

now any appropriate method  
 $x \approx 0.373$